

TOPICAL REVIEW

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Topical Review

Raman spectroscopic diagnostic of quantum spin liquids

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**Abstract**

Quantum spin liquids are outstanding examples of highly quantum entangled phases of matter and serve as a testbed to gauge central concepts of strongly correlated materials. Enormous research efforts in the past few decades have brought an in-depth understanding of these novel phases, although their conundrums have not yet been solved completely.

In this review, we give an overview of the three different classes of spin-liquid materials: (i) a one-dimensional spin chain system KCuF_3 , (ii) a kagome antiferromagnet $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$, and (iii) a Kitaev honeycomb material $\alpha\text{-RuCl}_3$. The emphasis is on demonstrating the success of the Raman scattering technique for probing fractionalized excitations in the aforementioned spin-liquid compounds, complementing a well-established neutron scattering method. Irrespective of dimensionality, spin topology, and spin-exchange type, the three materials share several common features in the spectral shape and temperature dependence of magnetic excitations, which can be taken as Raman spectroscopic fingerprints of quantum spin liquids.

Keywords: quantum spin liquid, Kitaev honeycomb lattice, Raman scattering, frustrated quantum magnet

(Some figures may appear in colour only in the online journal)

1. Introduction

Quantum spin liquids (QSLs)—a magnetic analog of the fractional quantum Hall liquid—epitomize key notions of modern condensed matter physics. These include fractionalization of elementary excitations, topological order, and emergent gauge fields [1–3]. From a magnetic phase viewpoint, QSLs are qualitatively different from conventional symmetry-broken states of matter as their ground state wavefunction is characterized by long-range quantum entanglement between local degrees of freedom. The topological character of QSLs is manifested in quasiparticle excitations carrying fractional quantum numbers and non-local statistical interactions for a special class of QSLs. Such quasiparticles cannot be created in a single isolated form but only in multiple pairs.

Broadly speaking, QSLs emerge in spin systems where the corresponding classical model has a macroscopic ground state degeneracy. Strong thermal and quantum fluctuations induced by the macroscopic degeneracy disrupt the development of long-range magnetic order. This approach has been pursued for geometrically frustrated magnets made of corner-sharing triangles or tetrahedra [1]. The canonical example in two dimensions is a nearest-neighbor (nn) Heisenberg model on the kagome lattice (=a lattice of corner-sharing triangles). In the kagome lattice, continuous rotations of spin clusters can be generated without incurring an energy cost, thereby leading to a destabilization of classical magnetic order [4]. Various numerical calculations of the kagome spin system unravel that a number of states including gapped $\mathbb{Z}_2$ and gapless $\text{U}(1)$ spin liquids are almost degenerate in their energies [2, 5, 6].

Currently, there is no general rigorous theory that allows for describing correlations, excitations, and topological properties of QSLs on the kagome lattice.

In this context, two exactly solvable spin models occupy a special position in the research of QSLs. The first one is a one-dimensional (1D) array of $s = 1/2$ spins that are exchange-coupled by nn antiferromagnetic Heisenberg interactions [7]. Its exact ground state forms a gapless algebraic QSL that corresponds to a macroscopic singlet entangling all spins in the chain. Its low-lying excitations are spinons that carry spin $s = 1/2$ and charge 0. The second one is a two-dimensional (2D) Kitaev honeycomb model $\mathcal{H}_K = K_\gamma S_i^\gamma S_j^\gamma$ ($\gamma = x, y, z$) where exchange on the three γ bonds of a honeycomb lattice involves the distinct Ising component of $S^\gamma S^\gamma$ [8, 9]. Its ground state is a Z_2 QSL. The excitation spectrum is composed of localized flux excitations with a tiny energy gap and itinerant Majorana fermions that are either gapped or gapless, depending on the ratio of K_γ . Even more salient is the emergence of anyonic quasiparticles in an external magnetic field that holds promise for fault-free topological quantum computation. In passing, we mention that QSLs in one dimension are distinct from those in higher dimensions in that braiding is not permitted in one dimension.

Generically, inelastic neutron scattering (INS) is regarded as a powerful tool for investigating multi-particle fractionalized excitations that are a dynamic fingerprint of QSLs. However, the neutron experiment is often hindered by the difficulties in synthesizing large-volume samples or obtaining a strong-enough scattering signal from many samples. In this vein, inelastic light scattering can be an experimental choice as the required sample size for this optical experiment is an order of hundred μm , and it yields a generally larger scattering intensity. As sketched in figure 1, Raman scattering allows probing magnetic collective excitations ranging from conventional spin waves to fractionalized spinons. More importantly, fermionic statistics in the fractionalized continuum can be directly measured by Raman scattering for a certain class of QSLs [10]. In table 1 we compare key features of Raman and neutron scattering techniques used to study magnetic excitations of matter.

In this Review, we present three case studies of the representative QSL candidates utilizing Raman spectroscopy. They are (i) the 1D spin chain system KCuF_3 , (ii) the kagome antiferromagnet $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$ (herbertsmithite), and (iii) the Kitaev honeycomb material $\alpha\text{-RuCl}_3$. Our objective is to make Raman spectroscopy an accessible method to the QSL community through a detailed comparison of neutron and Raman scattering data in the three compounds.

2. Principle of magnetic Raman scattering

A Raman spectrum of a magnetically ordered insulator usually consists of phonons and magnetic Raman scattering contributions. A distinction between both excitations is not always straightforward, yet some considerations provide useful guidance. For a given crystallographic structure, a factor group analysis yields a total number of Raman-active phonon modes

[11]. They can be identified based on their selection rules and characteristic spectral form. Since phonons arise mainly from a first-order scattering process, they have a well-defined Lorentzian lineshape and their linewidth is usually narrow ($2\text{--}10\text{ cm}^{-1}$). In sharp contrast, a two-magnon signal arises from a second-order scattering process and thus its spectral width is given by a magnon density-of-states (see figures 4 and 7 for the typical phonon and magnetic (color shadings) excitations). However, its specific lineshape is not described by a simple analytic function. In addition, tracing the temperature dependence can be a useful method for discriminating between magnetic and phononic Raman signals. When the sample is heated to room temperature, the phonon modes display a softening by several cm^{-1} and a small line broadening due to anharmonic phonon–phonon interactions. This is contrasted by the magnetic excitations that undergo substantial damping above the critical temperature (see figure 4(c) for a thermal evolution of the two-magnon spectrum). If no structural phase transition occurs at the magnetic ordering temperature, additional signals appearing below the ordering temperature can be tentatively assigned to magnetic signals. We further note that magnetic Raman scatterings have not always smaller light scattering cross-sections than phonons because they rely on their different microscopic mechanisms.

In the following, we briefly introduce Raman scattering in magnetic materials for the readers who are not familiar with the principle of a magnetic Raman scattering process. As sketched in figure 2(a), the incoming photon with frequency ω_i excites a virtual particle-hole pair, which in turn emits magnetic excitations (magnons, spinons,...) before it annihilates into an outgoing photon with frequency ω_f . Through optical processes, the magnetic excitations induce a spatially periodic modulation of the susceptibility tensor $\chi_{\alpha\beta}$. The modulation of $\chi_{\alpha\beta}$ can be expanded in Taylor series of spin operators $\mathbf{S}$. The perturbation terms give rise to one-magnon creation and annihilation [12]. In the microscopic mechanism, a single spin-flip occurs by the transition from a $S^z = S$ ground state to $S^z = S - 1$ through a successive electric dipole transition via the $L = 1$ virtual intermediate state in the presence of spin–orbit interaction $\lambda \mathbf{L} \cdot \mathbf{S}$.

In antiferromagnets and even antiferromagnetically correlated systems, two-magnon scattering is well established to arise from a fundamentally different process called exchange scattering mechanism [13]. This exchange mechanism is based on a double spin-flip caused by matrix elements of the Coulomb interaction conserving the total z component of spins ($\Delta_{\text{tot}}^z = 0$). In antiferromagnetically ordered states, this condition is fulfilled for the creation or annihilation of an even number of magnons. As a consequence, it usually has a larger scattering cross-section than one-magnon scattering.

Although Raman scattering in Mott insulators involves virtual charge excitations, the scattering process can be described effectively by a pure spin Hamiltonian within the Loudon–Fleury scheme [13–15]. The rationale behind the success of the Loudon–Fleury theory lies in the fact that in many cases photon frequencies are well below the charge gap and thus electronic degrees of freedom are frozen out. In a nutshell, given the nn Heisenberg interaction $\mathcal{H}_H = J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j$, the

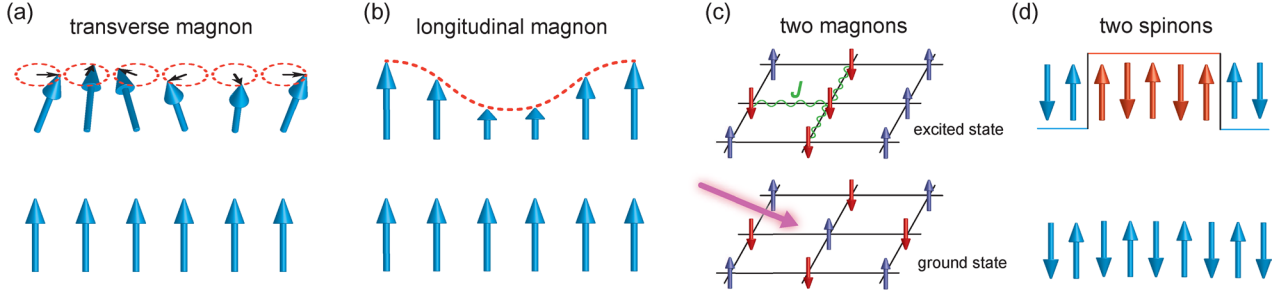


Figure 1. Raman scattering can probe (a) transverse magnons (=a single spin flip) in conventional magnets, (b) longitudinal magnons (=a periodic modulation of the magnitude of magnetic moment) in spin systems subject to strong quantum fluctuations, (c) two-magnon excitations in conventional antiferromagnets, and (d) spinons carrying $s = 1/2$ (=local spin flips surrounded by domain walls) in spin chains.

Table 1. Comparison between neutron and Raman scattering techniques used to probe magnetic excitations. Here, BZ is an abbreviation for Brillouin zone; SP for scattering process. $\mathcal{S}_{\text{tot}}$ represents the dynamic structure factor.

Property	Inelastic neutron scattering	Raman scattering
q -resolution	over an entire BZ	$q \approx 0$ for first-order SP; entire BZ for higher-order SP
Energy range	a few μeV – 20 eV	1 meV–1 eV
Energy resolution	5%–10% of incident neutron energies	<0.2 meV
Typical acquisition time per spectrum	(1) 3–5 h for powders (2) a few days for $\mathcal{S}_{\text{tot}}$ in single crystals	1–10 min
Sample volume	Several grams	μm -sized bulk and thin films
Magnetic field range	0–15 T	0–45 T
Selection rules	$\Delta S_z = \pm 1$	$\Delta S_z = 0$
Assignment of phonon and magnetic excitations	q dependence	Temperature and polarization dependence

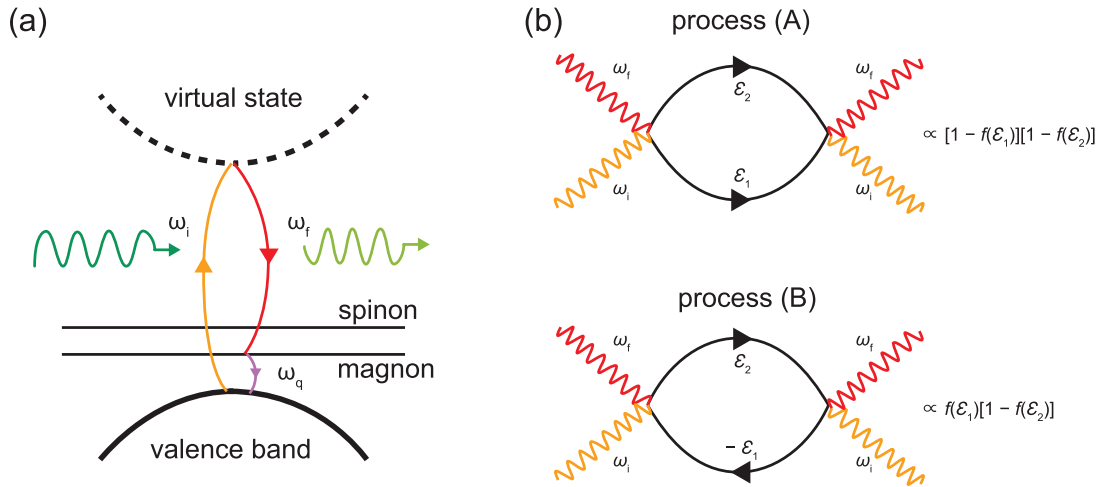


Figure 2. (a) Schematic of the Stokes Raman scattering process. (b) Feynman diagrams of the Raman scattering processes in Kitaev quantum spin liquids. Process (A) describes a creation or annihilation of a pair of itinerant Majorana fermions, while process (B) depicts a combination of creation and annihilation of itinerant Majorana fermions.

Loudon–Fleury operator for interaction between light and spin degrees of freedom gives

$$\mathcal{R} = \sum_{\langle ij \rangle} (\varepsilon_i \cdot \hat{r}_{ij}) (\varepsilon_s \cdot \hat{r}_{ij}) \mathbf{S}_i \cdot \mathbf{S}_j, \quad (1)$$

where ε_i and ε_s are the polarization vectors of the incident and scattered photons and $\hat{r}_{ij}$ is the unit vector connecting the sites i and j . Noteworthy is that the two-magnon scattering Hamiltonian has the same form as the Heisenberg Hamiltonian $\mathcal{H}_H$, since the Coulomb interaction is commonly involved in

the exchange mechanism. Using the Loudon–Fleury operator, one can obtain the Raman intensity as

$$I(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \mathcal{R}(t) \mathcal{R}(0) \rangle, \quad (2)$$

where $\mathcal{R}(t) = e^{i\mathcal{H}_H t} \mathcal{R} e^{-i\mathcal{H}_H t}$ and $\omega = \omega_f - \omega_i$ is the Raman shift.

In Kitaev QSLs, it is theoretically known that two different Raman processes contribute to the Raman spectrum [10]. In process (A), a photon scattering creates or

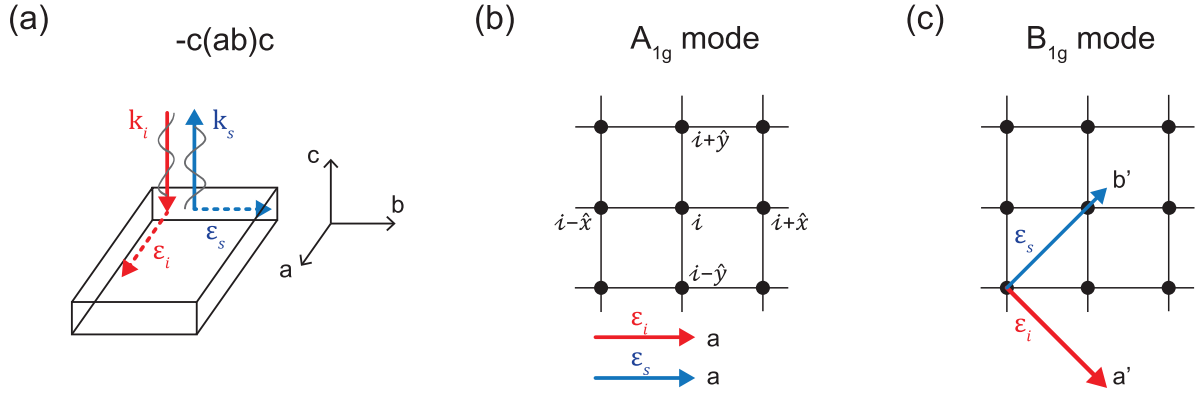


Figure 3. (a) Schematic representation of polarization-dependent Raman scattering measurements. k_i (ϵ_i) and k_s (ϵ_s) are the propagation (polarization) vectors of the incident and scattered wave vectors, respectively. The scattering configuration is denoted by the Porto notation $-c(ab)c$ with the crystallographic a , b , and c axes. A schematic picture of the incident and scattered light polarization in (b) the A_{1g} and (c) the B_{1g} experimental configurations on a square lattice. The arrows indicate the electric field vectors of the incident and scattered light.

annihilates a pair of itinerant Majorana fermions with energies ϵ_1 and ϵ_2 (see figure 2(b)). Applying the Loudon–Fleury theory of equation (1) to $\mathcal{H}_K$ (introduced in section 1), one obtains an asymptotic form of the Raman intensity as $I_A(\omega) \propto [1 - f(\epsilon_1)][1 - f(\epsilon_2)]\delta(\omega - \epsilon_1 - \epsilon_2)$ with the Fermi distribution function $f(\omega) = 1/(1 + e^{\hbar\omega/k_B T})$. In process (B), the scattering creates a fermion with energy ϵ_2 and annihilates a fermion with energy ϵ_1 simultaneously (see figure 2(b)). The temperature dependence of the Raman intensity is proportional to $I_B(\omega) \propto f(\epsilon_1)[1 - f(\epsilon_2)]\delta(\omega + \epsilon_1 - \epsilon_2)$. Thus, Raman scattering data provide a measure of identifying fractional Majorana excitations as their occupancy in the flux background obeys Fermi statistics.

In magnetic insulators, phonon and magnetic excitations depend on the direction and polarization of the incident and scattered light with respect to the crystallographic axes. Generically, the scattering configurations are denoted by the Porto notation, namely, $k_i(\epsilon_i, \epsilon_s)k_s$ [16]. Here the labels outside the bracket are the propagation vectors of the incident (k_i) and scattered (k_s) light. ϵ_i and ϵ_s are the polarization vectors of the incoming and detected light, respectively. Figure 3(a) sketches a schematic representation for polarization-dependent measurements. The crystallographic axes a and b lie in the ab plane and the c axis is perpendicular to the ab plane. In backscattering geometry, we have $k_i = -k_s$ and, thus, the scattering configuration is denoted by $-c(ab)c$ or $-z(xy)z$. Throughout this paper, we will use a shorthand notation $(\epsilon_i, \epsilon_s) = (ab)$.

We next consider the two scattering geometries for a Heisenberg square-lattice model with a tetragonal (D_{4h}) group shown in figures 3(b) and (c). In the first measurement geometry, (ϵ_i, ϵ_s) both are parallel to the crystallographic a axis, while the ab plane of the crystal is perpendicular to the wave vector k_i of the incident beam. This geometry probes modes of A_{1g} symmetry. In the second ($a'b'$) geometry, (ϵ_i, ϵ_s) both are rotated from the a and b axes by 45° . B_{1g} modes are observed in this scattering configuration. The A_{1g} and B_{1g} modes are classified according to whether $\chi_{\alpha\beta}$ is positive or negative with respect to 90° rotation about the principal c axis. The subscript g stands for *gerade* (even) under the inversion. Now, we calculate the Loudon–Fleury operator for the A_{1g}

configuration. In the square lattice, there are the four nn unit vectors $\hat{r}_{ij} = \{\hat{x}, -\hat{x}, \hat{y}, -\hat{y}\}$ as depicted in figure 3(b). With $\epsilon_i = \epsilon_s = \hat{x}$, equation (1) becomes

$$\mathcal{R}_{A_{1g}} = \sum_{\langle ij \rangle, x} \mathbf{S}_i \cdot \mathbf{S}_j + \sum_{\langle ij \rangle, y} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (3)$$

where the first (second) sum is over the nn bonds along the x (y) axis. This Raman operator is nothing but the Heisenberg Hamiltonian $\mathcal{H}_H$. Since $[\mathcal{R}_{A_{1g}}, \mathcal{H}_H] = 0$, the theory predicts no scattering in the A_{1g} scattering geometry. In the B_{1g} configuration, with $\epsilon_i = \frac{1}{\sqrt{2}}(\hat{x} - \hat{y})$ and $\epsilon_s = \frac{1}{\sqrt{2}}(\hat{x} + \hat{y})$, we obtain the Raman operator

$$\mathcal{R}_{B_{1g}} = \sum_{\langle ij \rangle, x} \mathbf{S}_i \cdot \mathbf{S}_j - \sum_{\langle ij \rangle, y} \mathbf{S}_i \cdot \mathbf{S}_j. \quad (4)$$

Since $[\mathcal{R}_{B_{1g}}, \mathcal{H}_H] \neq 0$, there will be a finite scattering intensity for the B_{1g} geometry. For the calculations of a phonon scattering intensity, we refer to literature [17].

As an example of two-magnon scattering we present the experimental Raman spectra of the 2D antiferromagnet LaSrMnO_4 ($3d^4; s=2$) in figure 4. This compound shows the Néel ordering at $T_N = 133$ K and is a good realization of the Heisenberg square-lattice model with a tetragonal (D_{4h}) group [18]. According to the selection rule, two-magnon scattering is allowed in the B_{1g} scattering geometry. In ($a'b'$) polarization, indeed, we observe the well-defined maximum at 356 cm^{-1} (blue shaded excitation in the 5 K spectrum) assigned to two-magnon excitation that corresponds to a double spin-flip process of a ground state ($S^z = \pm 2$) to a higher state ($S^z = \pm 1$) (see the sketch of figure 1(c)). Upon heating through T_N , the maximum broadens and shifts to lower energies and finally evolves to quasielastic scattering at high temperatures (red shaded area). The thermal evolution of the two-magnon spectrum is largely dictated by the temperature dependence of short-wavelength magnon (=a magnon near the boundary of the Brillouin zone) energies and lifetimes because the two-magnon density of states is largest at the zone boundary and long-wavelength magnons renormalize more rapidly with temperature than short-wavelength magnons. Consequently, the peak energy and linewidth of the two-magnon spectrum mainly reflect the short-wavelength magnon energy and

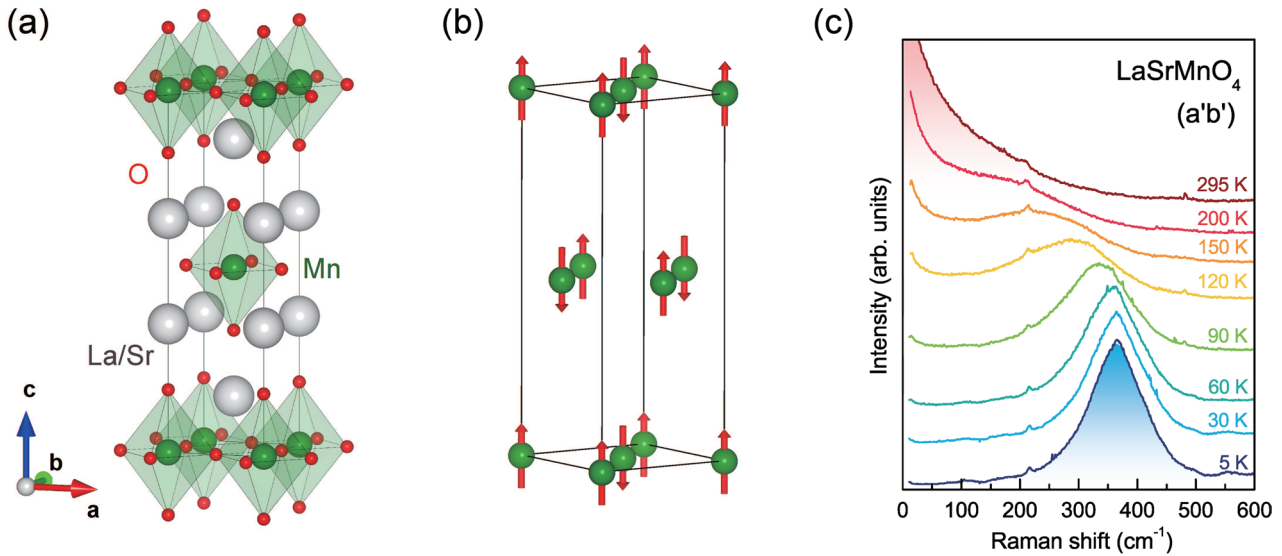


Figure 4. (a) Crystal structure of LaSrMnO₄. (b) Magnetic ordering pattern of LaSrMnO₄ below the ordering temperature $T_N = 133$ K. (c) Thermal evolution of two-magnon scattering in $a'b'$ polarization (blue shading). The red shading highlights the quasi-elastic scattering at $T = 295$ K. Reprinted with permission from [18], © 2008 American Physical Society.

lifetime, respectively. At elevated temperatures above T_N , say 200 K, we still see the hump structure around 200 cm^{-1} , coexisting with the quasielastic scattering. This means that short-range magnetic fluctuations survive well above T_N . At 295 K ($\approx 2.2 T_N$), the magnetic Raman scattering is approximated by the quasielastic response with Lorentzian-like shape. This incoherent magnetic scattering arises from spin diffusion or spin-energy fluctuations [19]. In the classical antiferromagnets, the peak energy E_{2M} of the two-magnon scattering allows the estimation of the exchange coupling constant J by the relation $E_{2M} = J(2zs - 1)$ where z is the number of nn spins [12]. For LaSrMnO₄, we obtain the value of $J \approx 34$ K with $s = 2$ and $z = 4$. This example highlights Raman scattering as a valuable tool for determining magnetic exchange interactions.

3. 1D $s = 1/2$ chain KCuF₃

Pure 1D chains with solely nn intrachain couplings evade any long-range order [20]. Instead, regions of short-range correlations are separated by topological defects—spinons that act as $s = 1/2$ fractional excitations. However, small deviations from this purely 1D model such as minute interchain couplings lead to the stabilization of conventional magnetic order at the lowest temperature. KCuF₃ is a close experimental realization of a 1D $s = 1/2$ chain system with antiferromagnetic intrachain coupling. As shown in figure 5(a), the Cu²⁺ ions are octahedrally coordinated by F⁻ ions and form chains along the crystallographic c axis. From the static magnetic susceptibility in figure 5(b), the antiferromagnetic exchange coupling of $J = 190$ K (≈ 16.5 meV) was estimated [21]. The presence of additional interchain interactions results in magnetic order below $T_N = 39$ K. At high temperatures ($T_{OO} = 800$ K) orbital order sets in, and a freezing of dynamical distortions concerning the apical F⁻ ions that connect the octahedra along c is observed around $T_S = 50$ K [22].

Raman scattering is a particularly relevant probe for quasi-1D spin chains such as KCuF₃ as it simultaneously probes lattice and spin degrees of freedom, and via selection rules (light polarization vectors ε_i and ε_s with respect to $\hat{r}_{ij}$) can easily distinguish between intrachain and interchain contributions. A thorough investigation of the phonon modes revealed a symmetry lowering of the crystal structure as a consequence of both a transition from dynamic to static distortions through $T_S = 50$ K and the magnetic ordering through $T_N = 39$ K [23]. A clear indication for a lowered symmetry is the observed splitting of a phonon mode below T_S (see figure 5(c)). In addition, an anomalous broadening of a phonon mode, different from conventional anharmonic broadening, is observed for several modes for $T > T_S$. This broadening can be mapped surprisingly well onto the linewidth broadening observed via ESR [24] (see figure 5(d)), underlining an intricate interplay of lattice and orbital degrees of freedom in KCuF₃ [25].

In figure 6 we present the magnetic excitation spectrum of KCuF₃ probed by INS experiments. The cartoon of spinon excitations is sketched in figure 6(a): short-range spin correlations in 1D are broken up by topological defects that are created in pairs (red spins). A continuum of spinon excitations of a 1D $s = 1/2$ Heisenberg antiferromagnet is depicted in figure 6(b). Its lower and upper bounds as described by des Cloizeaux & Pearson [26] as well as by Faddeev & Takhtajan [27]: $\omega_{\text{lower}}(k) = \frac{\pi}{2}J|\sin(k)|$ and $\omega_{\text{upper}}(k) = \pi J|\sin(k/2)|$. To a good approximation, such a spinon continuum is present in KCuF₃ as revealed by the INS data (see figures 6(c)–(e)) [28, 29]. Nevertheless, small deviations arising from finite interchain couplings modify an excitation spectrum at the lower boundary (below 25 meV). In particular, the signature of a longitudinal magnon mode is present in the neutron data below T_N (see figure 6(f)), where an enhanced scattering intensity is observed above 15 meV. This conventional magnetic excitation marks the presence of long-range magnetic order subject to strong quantum fluctuations. The sudden high-energy

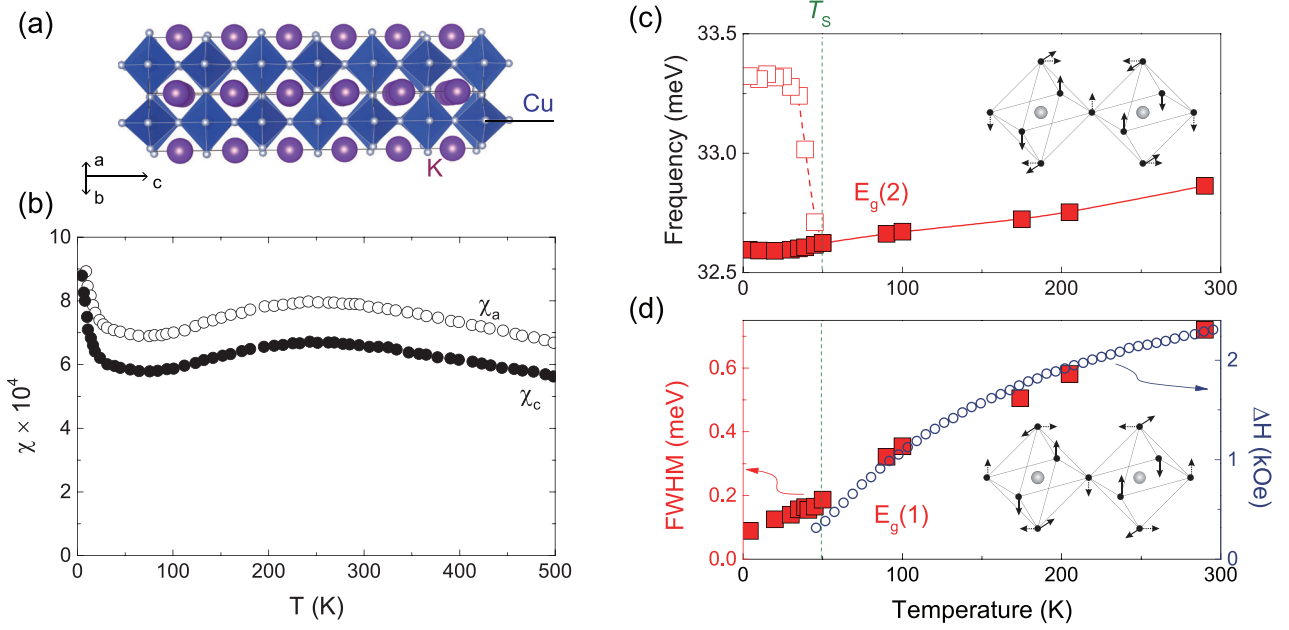


Figure 5. (a) Cu^{2+} ions octahedrally coordinated by F^- ions forming chains along the c direction. (b) Temperature dependence of the magnetic susceptibility for $H||a$ and $H||c$. Reproduced with permission from [21]. (c) Temperature dependence of the $E_g(2)$ phonon frequency. The phonon splitting marks the symmetry lowering below T_s . (d) Anomalous phonon broadening at $T > T_s$ observed for the $E_g(1)$ phonon (red squares) and a comparison to the linewidth broadening of an ESR absorption line (blue circles). The insets depict ionic displacements for the $E_g(1)$ and $E_g(2)$ modes. Reprinted with permission from [24], © 2008 American Physical Society.

cut-off above 80 meV is ascribed to coupling to lattice degrees of freedom.

Next, we will turn to magnetic excitations probed by Raman spectroscopy and compare them to the neutron data. Conventional magnetic excitations shown in figures 1(b)–(c) can be observed in the antiferromagnetic phase of KCuF_3 . Below T_N two new modes emerge with very specific selection rules: in bc polarization (i.e. with the scattered light polarization vector ε_s along the chain direction, namely, the crystalline c axis) a very weak but well-defined excitation is seen that hardens with decreasing temperature and saturates at around 14 meV (figure 7(a)). A second excitation with a larger linewidth and stronger scattering intensity is seen in aa and bb polarizations, i.e. with the incident and scattered light polarized perpendicular to the chain direction [$\varepsilon_i, \varepsilon_s \perp c$, figure 7(b)]. Its energy at base temperature is around 18 meV. Here we note that no magnetic scattering is allowed in the intrachain cc polarization because the Loudon–Fleury operator in equation (1) commutes with the spin chain Hamiltonian. Based on their selection rules and characteristic energies, these signals are assigned to a transverse magnon and a longitudinal magnon, respectively. The former mode can be described by classical spin-wave theory, hence it arises from the presence of (weak but) finite interchain coupling. The longitudinal mode emerges as a crossover characteristics from a 3D-dominated system (at low temperatures $< T_N$ and low energies < 12 meV) to a 1D-dominated system (at high temperatures and high energies > 25 meV) and requires both long-range order as well as a certain amount of quantum fluctuations [29, 30].

Shown in figure 7(c) is the $T = 3.5$ K Raman spectrum in bb polarization. We observe a broad background (blue shading) that extends up to ~ 80 meV in addition to the sharp

phonon modes ($E_g(1)$ and $E_g(2)$ are marked), a quasielastic tail centered around 0 meV due to magnetic energy fluctuations [31] (red-shaded contribution), and the longitudinal mode (shaded in orange). This continuum dominates the high energy range above 25 meV, while its contribution at lower energies is somewhat ambiguous due to the overlap with the quasielastic tail and the longitudinal mode. This boundary (dashed red line) marks the crossover from a 1D Luttinger liquid to a 3D nonlinear σ physics. Remarkably, the energy range of this continuum is in nice agreement with the INS data of the spinon continuum presented in figure 6(e), which has an upper cut-off around 80 meV. Further, we stress that the coexistence of the longitudinal magnon and the diffusive scattering constitutes the landscape of low-energy excitations in the 3D nonlinear σ model region below 25 meV.

4. 2D kagome lattice antiferromagnet $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$

When realizing a QSL ground state, several ingredients are desirable: low dimensionality D , low coordination z , and a small quantum spin number s . To that end, a 2D $s = 1/2$ antiferromagnet on the kagome lattice serves as a promising candidate. The term kagome stems from the traditional Japanese art of basket weaving and consists of a network of corner-shared triangles as shown in figure 8(a).

Among the reported compounds, $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$ known as ‘herbertsmithite’ is closest to a perfect kagome lattice in that it consists of isosceles triangles and thereby has a maximum degree of spin frustration. Its 2D kagome planes are composed of the Cu^{2+} ions ($s = 1/2$) with

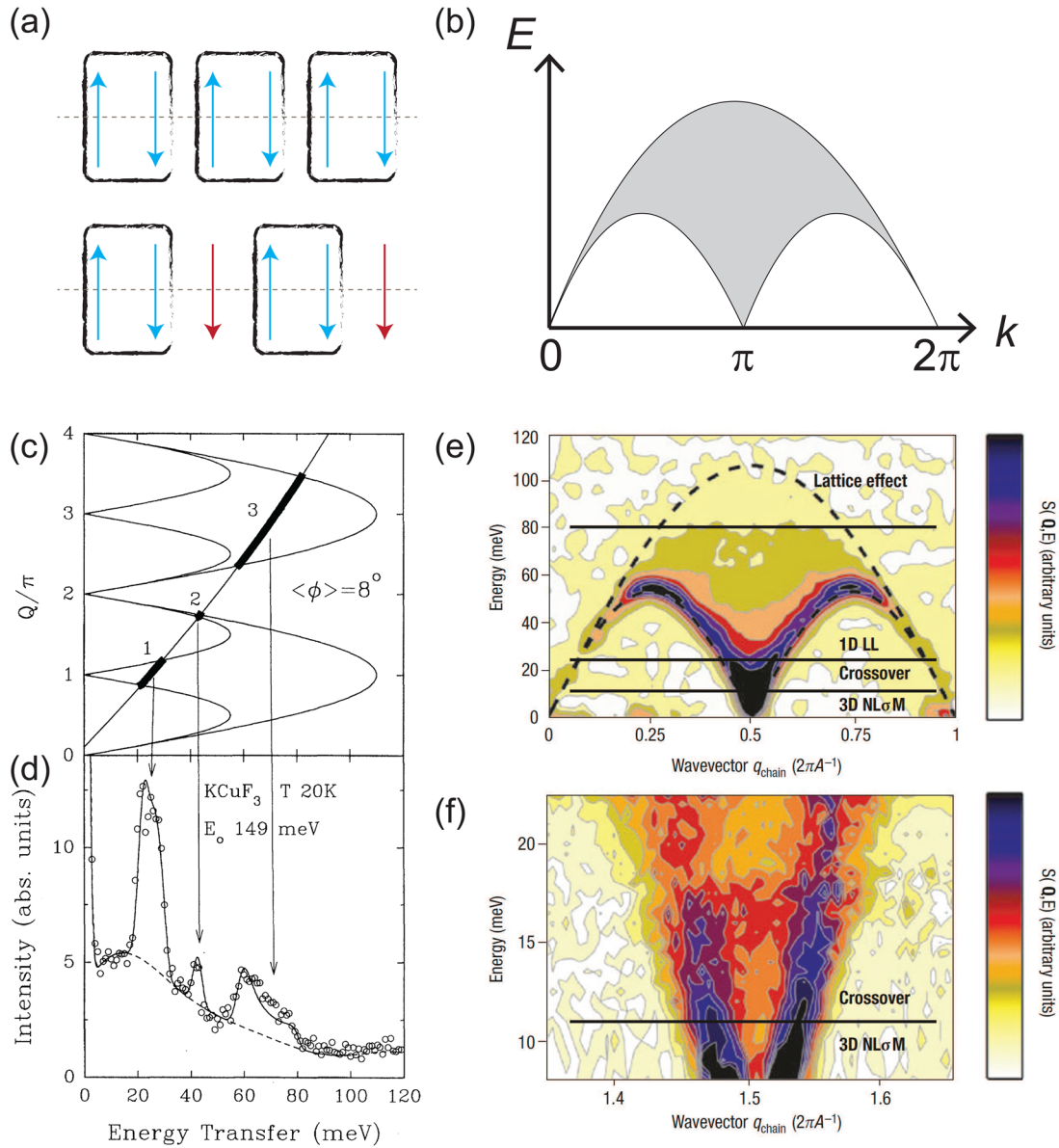


Figure 6. (a) Creation of a pair of $s = 1/2$ spinons in a 1D chain. (b) Spinon continuum modeled after des Cloizeaux & Pearson [26] and Faddeev & Takhtajan [27]. (c) Continuum spanned by spinon excitations. The solid line highlights regions that contribute to the spectral weight at the given experimental condition. (d) Inelastic neutron scattering data of KCuF_3 at $T = 20$ K. Reprinted with permission from [28], © 1993 American Physical Society. (e) Energy dispersion of the spinon continuum measured by inelastic neutron scattering at $T = 6$ K. (f) Zoom into the cross-over regime with an enhanced spectral weight due to the longitudinal magnon. [29] © 2005 Macmillan Publishers Limited. With permission of Springer.

magnetic superexchange mediated via hydroxyl-groups (see figure 8(b)). The antiferromagnetic exchange coupling is $J \approx 195$ K ($=17$ meV). Yet, thermodynamic measurements found no evidence of long-range order even down to 50 mK [32]. Meanwhile, $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$ is subject to intersite mixing between the Cu^{2+} ions and non-magnetic Zn ions. This can effectively dilute the kagome layers while establishing a finite interlayer coupling. Kagome lattices with a classical spin (i.e. $s > 1/2$) were shown to realize weathervane modes as concerted rotations of spins [33]. The ground state of a kagome antiferromagnet in the quantum spin limit ($s = 1/2$), on the other hand, is predicted to have a resonating valence bond (RVB) state—a highly entangled state without any magnetic order [34]. Here, the lattice is covered by dimer bonds with

a macroscopically large degeneracy of possible patterns as sketched in figure 8(c). In other words, the ground state can be visualized as a superposition of each possible pattern in analogy to Pauling's valence bond description of the benzene ring. An excitation from this ground state would correspond simply to the breaking of one dimer bond, thus creating two free spins that can propagate in the sea of RVBs with vanishingly small energy. Therefore, they can be considered as two individual excitations, i.e. fractional ($s = 1/2$) spinon excitations.

The T dependence of the magnetic susceptibility of $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$ is shown in figure 9(a) with no sign of long-range magnetic order [35]. As seen from figure 9(b), the $T = 1.6$ K INS data of the deuterated single crystals show a diffuse magnetic scattering without any well-defined peak

in momentum space [36]. This magnetic response is in contrast to conventional spin-wave excitations and can instead be theoretically modeled in terms of uncorrelated nn dimers (see figure 9(c)). In figure 9(d) we plot the energy dependence of the scattering function $S_{\text{tot}}(Q, \omega)$ at various points in reciprocal space. No sign of an excitation gap is observed down to at least $\hbar\omega = 0.25$ meV. $S_{\text{tot}}(Q, \omega)$ is largely constant over temperature and forms a broad continuum up to the highest measured energy of 11 meV. This continuum accounts for about 20 % of the total magnetic moment, suggesting that it extends far beyond 11 meV. A rough extrapolation yields an upper limit of the continuum around 50–60 meV.

As-measured Raman spectra of $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$ are shown in figure 10(a) at various temperatures. Here, the incident laser beam is perpendicular to the kagome planes such that the $\vec{E}$ -field of the light is within the planes. Note that the spectra are shifted in intensity for clarity. From the phonon modes, we find no hint of a structural phase transition as a function of temperature. Therefore, we can assume the preservation of structurally perfect 2D kagome layers down to the base temperature. Unlike the 1D spin chain system KCuF_3 , there appear no conventional magnetic excitations associated with long-range magnetic order. As the temperature decreases, a broad and featureless background emerges over a wide energy range. In figure 10(b) the spectra at $T = 295$ K and 5 K are replotted after subtraction of phonon modes as well as Bose correction to remove thermal influence. The broad continuum becomes clearly visible. We find that no onset energy exists and the intensity is inclined to drop to 0 as $\hbar\omega \rightarrow 0$, pointing towards a gapless continuum, in full agreement with the INS data. The upper limit of the continuum lies around 60–70 meV ($\approx 4J$). This is also fully consistent with the INS results, where the rough extrapolation of the continuum suggested an upper limit of around 50–60 meV.

Upon heating the sample, the continuum gradually renormalizes towards smaller energies and eventually turns into a quasielastic tail that persists to room temperature. These observations are in stark contrast to conventional two-magnon scattering in long-range ordered states, where magnetic excitations yield a more well-defined signal with clear onset energy and a moderate renormalization at best (see figure 4(c)). Instead, the spectral shape and energy range of the continuum at base temperature can be approximated by theory based on the Shastry-Shraiman formulation under the assumption of a QSL ground state [37]. In particular, it was pointed out that an algebraic spin liquid should prevail in $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$, i.e. a state in which dynamic short-range magnetic correlations decay algebraically both spatially and temporally. A measure for such fluctuations is the quasielastic scattering intensity I_{QES} (see figure 10(a)). This quantity is directly related to the magnetic specific heat C_m via the relation $I_{\text{QES}} \sim C_m T^2$ [19]. As the temperature is lowered, figure 10(c) exhibits a power-law decay of I_{QES} , which is a fingerprint for algebraic correlations.

We also address the effect of impurities and intersite mixing on the ground state for two different samples: a naturally grown specimen retrieved from a chilenian copper mine (open squares), and a single crystal grown via hydrothermal

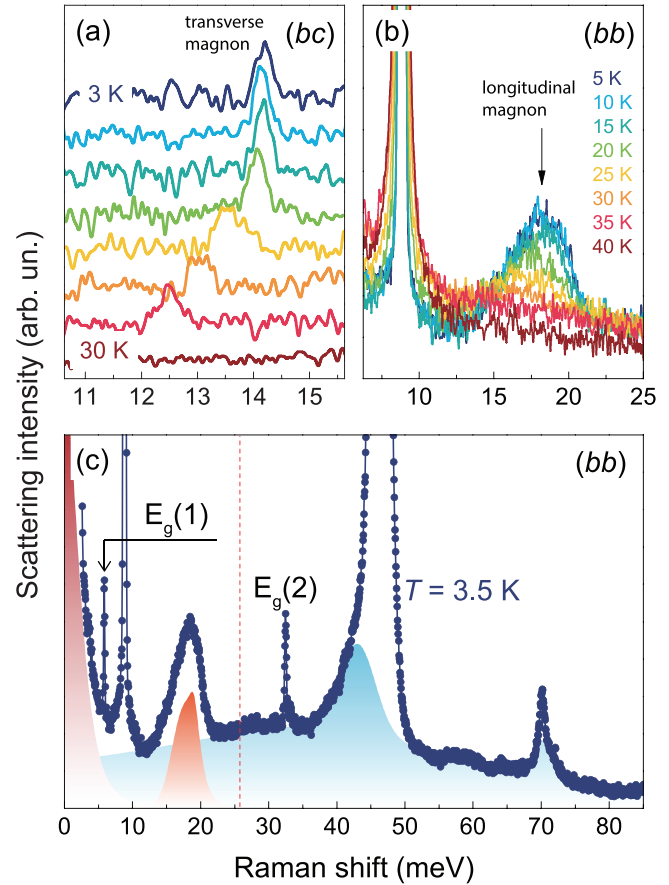


Figure 7. (a) Transverse magnon mode measured in *bc* polarization in the antiferromagnetic state below T_N . (b) Longitudinal magnon seen in *bb* polarization. (c) $T = 3.5$ K Raman data in *bb* polarization covering an extended energy range. A broad spinon continuum (blue shaded region) coexists with the longitudinal magnon (orange) and a quasielastic tail (red). Reproduced from [25], with the permission of AIP Publishing.

synthesis (open circles). The natural mineral suffers from a considerable amount of defects and impurities, and its stoichiometry is closer to $\text{Zn}_{0.8}\text{Cu}_{3.2}(\text{OH})_6\text{Cl}_2$. On the other hand, the stoichiometry of the synthesized sample is very close to $\text{Zn}_1\text{Cu}_3(\text{OH})_6\text{Cl}_2$ [38, 39]. As evident from figure 10(c), both compounds display an algebraic slowing of fluctuations with decreasing temperature. However, it is much more pronounced in the mineral sample. This points to the detrimental effect of impurities on the low-energy physics. This means that in the presence of impurities and intersite mixing, quantum fluctuations induced by frustration are quenched more effectively. On the contrary, the overall shape, scattering intensity, and energy range of the spinon continuum are remarkably robust against impurities, and both compounds show very similar Raman responses at base temperature [38].

Finally, Raman scattering is utilized to study the effect of structural distortions on the ground state of a 2D kagome lattice antiferromagnet [40]. Besides herbertsmithite, promising kagome lattice antiferromagnets include vesignieite and volborthite. Both $s = 1/2$ compounds have slightly distorted 2D kagome layers (see figure 11 for details), resulting in $J_1 - J_2$ anisotropies. Vesignieite is made of nearly isosceles triangles with anisotropy of around 0.07 % [41]. In contrast

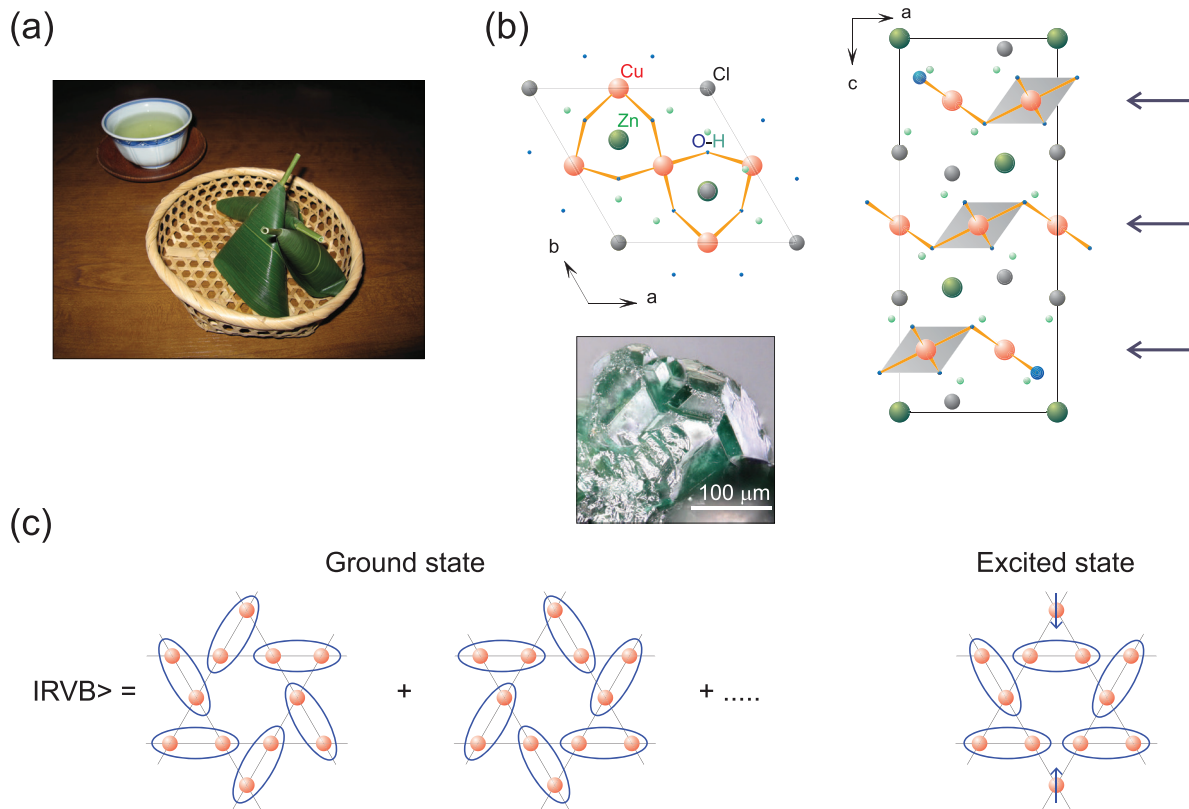


Figure 8. (a) A traditionally-woven Japanese basket with a kagome pattern. (b) Crystal structure of $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$. The 2D kagome layers within the crystallographic ab plane are marked by blue arrows. The photo shows a microscope image of the measured natural single crystal. (c) Schematic representation of a resonating valence bond state on a 2D kagome lattice (left). Spinon excitations from the resonating valence bond (right).

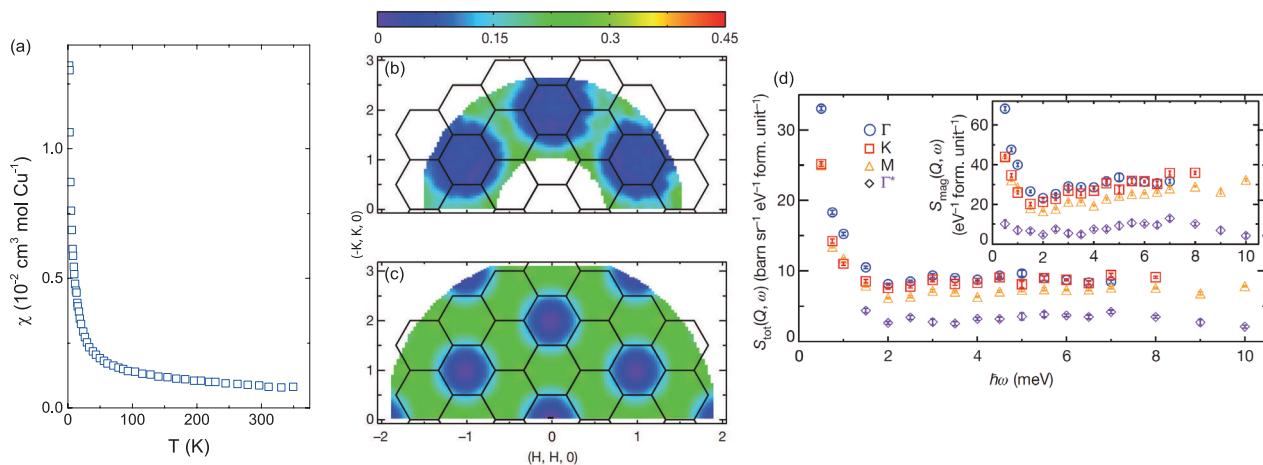


Figure 9. (a) Temperature dependence of the magnetic susceptibility of $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$. (b) Dynamic structure factor obtained at $T = 1.6 \text{ K}$ over an integrated energy regime $\hbar\omega = 1-9 \text{ meV}$. (c) Corresponding calculated structure factor. (d) Dynamic structure factor S_{tot} over energy measured at high symmetry points. The inset displays the solely magnetic contribution $S_{\text{mag}}(\mathbf{Q}, \omega)$. [36] © 2012 Macmillan Publishers Limited. With permission of Springer.

to herbertsmithite, the static magnetic susceptibility shows a broad maximum at about 22 K, evidencing the development of short-range correlations at low temperatures [42]. The distortion in volborthite is more severe, leading to anisotropy of around or slightly less than 20 % [43]. Here, the dynamic structure factor obtained by INS experiments hinted towards the onset of nn correlations below 50 K [44]. The Raman spectra of both kagome antiferromagnets are fundamentally

different from the broad, gapless spinon continuum observed in herbertsmithite. In vesignieite, a more well-defined excitation emerges around 115 cm^{-1} with clear onset- and cut-off energies. Its temperature dependence resembles characteristic features of the magnetic susceptibility, therefore pointing towards a magnetic origin. By applying the simple counting argument for a conventional two-magnon excitation, we estimate its associated magnetic exchange interaction to be

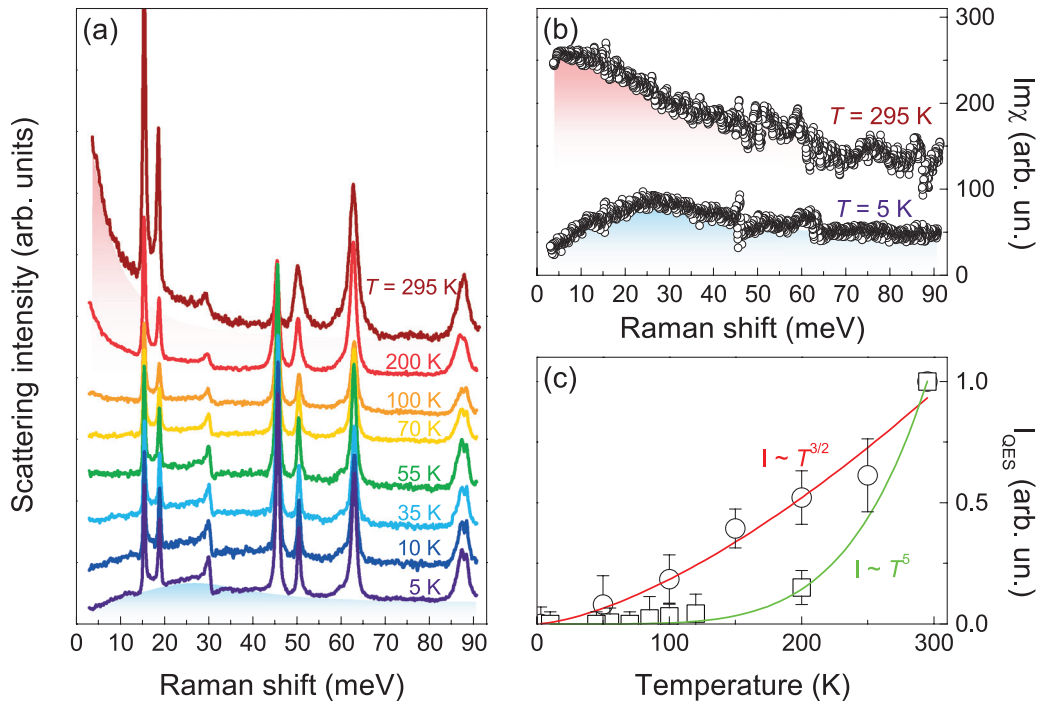


Figure 10. (a) Temperature-dependent as-measured Raman spectra of $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$. A broad continuum of fractional spinon excitations emerges towards low temperatures (blue shading). (b) Phonon-subtracted and Bose-corrected Raman data at $T = 295$ K and 5 K. The shadings denote the residual magnetic Raman response. (c) Thermal evolution of the quasielastic intensity (red-shaded area in panel (b)) for a natural specimen (open squares) and a synthesized single crystal (open circles). Reprinted with permission from [38] © 2010 by the American Physical Society.

around 55 K, which is in very good agreement with $J_{\text{vesi}} = 53$ K. In volborthite, the onset of nn correlations leads to an increase in distortion, such that the symmetry is further lowered and the spin frustration is relieved. This is evidenced by the emergence of numerous new phonon modes below 50 K [40]. Even, it has been proposed that a square lattice of trimers having effective $s = 1/2$ moments emerges from the original distorted kagome lattice [45]. The comparison of all three kagome compounds, as summarized in figure 11, impressively demonstrates the considerable impact that lattice distortions have on the ground state of a kagome lattice antiferromagnet. With increasing degree of lattice distortions, the deconfined spin excitations are confined and eventually condensed to magnons. As to the chemical disorders and impurities, the high-energy QSL state remains robust, while the low-energy physics is largely affected.

5. Kitaev honeycomb magnet $\alpha\text{-RuCl}_3$

Spin-orbit Mott insulators offer a promising platform for the celebrated Kitaev model due to an interplay of strong spin-orbit coupling and electron interactions [46]. Jackeli and Khaliullin proposed the designing principles of realizing bond-directional anisotropic Kitaev interactions in real materials [47]: (i) a Kramers doublet with the pseudospin $j_{\text{eff}} = 1/2$ under the cubic crystalline electric field and the spin-orbit coupling and (ii) quantum interference between the indirect electron hoppings through two different 90° exchange paths. These requisites are largely fulfilled in $4d$ and $5d$ transition metal compounds with the low-spin d^5 configuration

and the edge-sharing octahedral environments. The prototypical examples are a family of honeycomb iridates A_2IrO_3 ($\text{A} = \text{Li, Na}$) and a ruthenium trichloride $\alpha\text{-RuCl}_3$ [48, 49].

As established by extensive theoretical and experimental studies [3], these materials possess dominant ferromagnetic Kitaev interactions. However, other subdominant interactions give rise to the low- T magnetic order instead of the anticipated QSL ground state. The additional interactions, hardly avoided in materials, include symmetric anisotropic Γ , nn Heisenberg J , third nn Heisenberg J_3 , and anisotropic Γ' interactions. Notwithstanding, finite-temperature and -field signatures of the Kitaev QSL can be still attained in a quantum paramagnetic state as a proximity effect of a QSL state.

The elementary excitations of the Kitaev QSL arise from the fractionalization of spins. Spins are thermally fractionalized into two kinds of Majorana quasiparticles. The first ones are localized Majorana fermions called fluxes whose excitations are gapped with a small energy gap and form a Q -independent flat band. The second ones constitute itinerant Majorana fermions whose spectrum has a broad continuum ranging continuously from zero energy to high energy of an order of K . Thanks to the exact solvability of the Kitaev model, thermodynamic and spectroscopic quantities can be computed over almost the full temperature range [50–53]. In specific heat, the two fractional quasiparticles with distinct energy scales show up as two well-separated peaks. The high- T peak corresponds to the itinerant Majorana fermions, and the low- T peak is ascribed to a flux condensation of the localized Majorana fermions. Each peak carries an entropy of 50 % of $R \ln 2$ (R : ideal gas constant) of the $s = 1/2$ system. The dynamic spin structure factor $\mathcal{S}_{\text{mag}}(Q, \omega)$ captures the

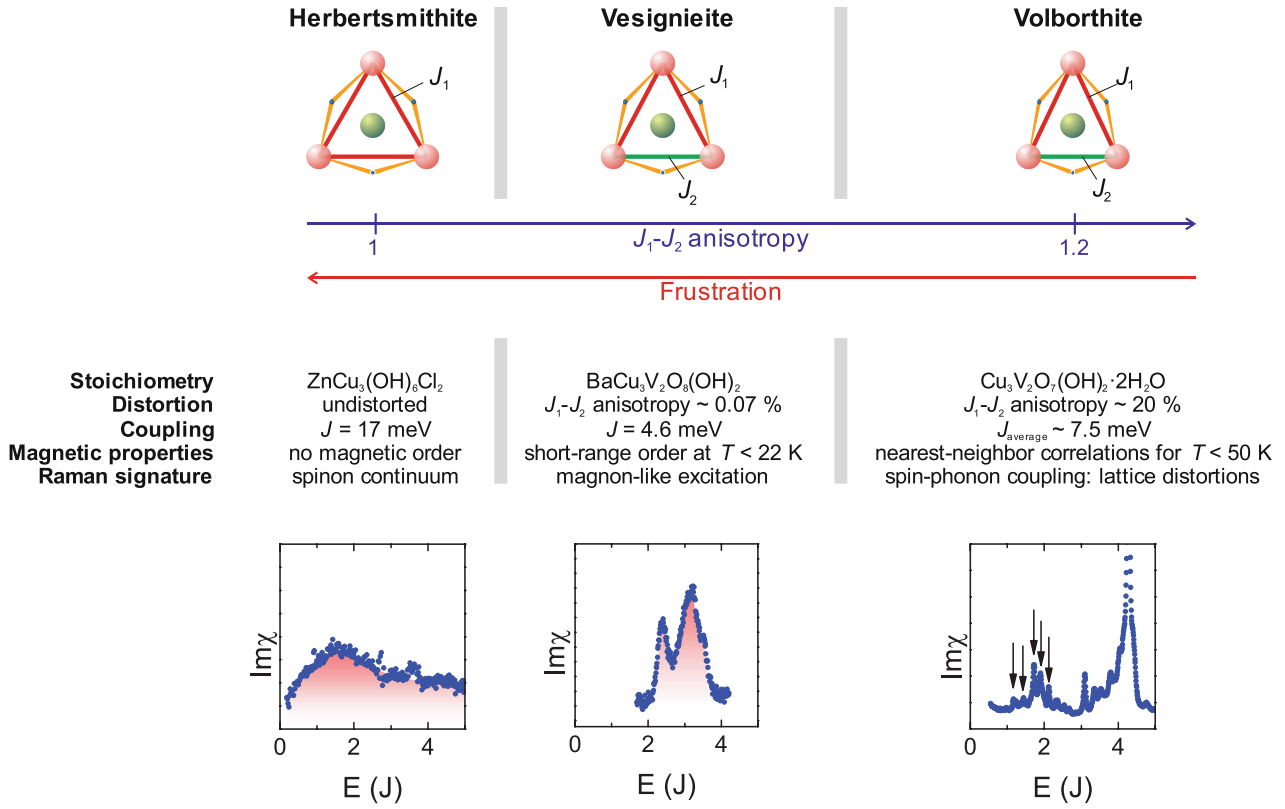


Figure 11. Effect of lattice distortion and the resulting $J_1 - J_2$ anisotropy on the physics of the kagome lattice antiferromagnets.

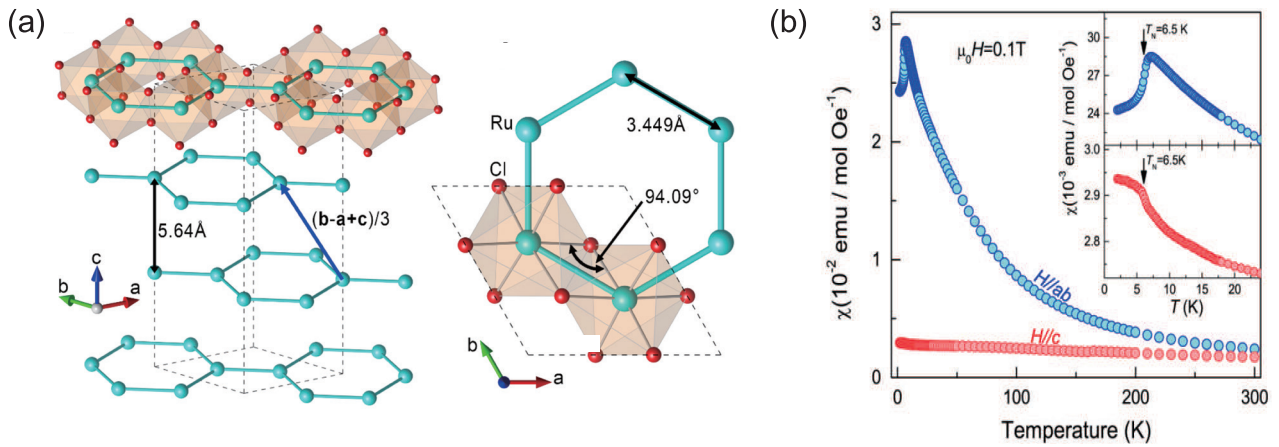


Figure 12. (a) Crystal structure of α - RuCl_3 with the Ru-Cl-Ru bond angle and distance [57]. (b) Temperature dependence of the static magnetic susceptibility for $H \parallel ab$ and $H \parallel c$. The inset is a zoom of the low- T magnetic susceptibility with an indication of magnetic order at $T_N = 6.5$ K.

two fermionic excitations, while the Raman response selectively probes the itinerant Majorana fermions. As discussed in section 1, the Raman scattering data can test whether the fractionalized excitations acquire a Majorana flavor by their temperature dependence.

In the past, Raman studies have been conducted on Kitaev candidates such as β - and γ - Li_2IrO_3 [54], $(\text{Na},\text{Li})_2\text{IrO}_3$ [55], as well as $\text{Na}_4\text{Ir}_3\text{O}_8$ [56]. However, insufficient sample sizes prevent a comparison between Raman and neutron scattering results. In the following, we will turn to α - RuCl_3 that forms a honeycomb network of edge-sharing RuCl_6 octahedra as shown in figure 12(a). Due to a small trigonal distortion

of the RuCl_6 octahedra, the Ru-Cl-Ru angle becomes larger than 90° (nearly 94°), giving rise to non-Kitaev terms [57]. Figure 12(b) presents the T -dependence of the static magnetic susceptibility $\chi(T)$ which shows a pronounced XY-like anisotropy between $H \parallel ab$ (in-plane) and $H \parallel c$ (out-of-plane). Given that $\chi(T)$ of the isotropic Kitaev model exhibits an isotropic behavior, the field-orientation-dependent anisotropy of $\chi(T)$ in α - RuCl_3 suggests the presence of bond-dependent anisotropic magnetic couplings such as off-diagonal Γ interaction. From the kink in the low- T $\chi(T)$, a zigzag magnetic ordering is identified at $T_N = 6.5$ K. The K- Γ model serves as the very minimal spin Hamiltonian that can explain the zigzag

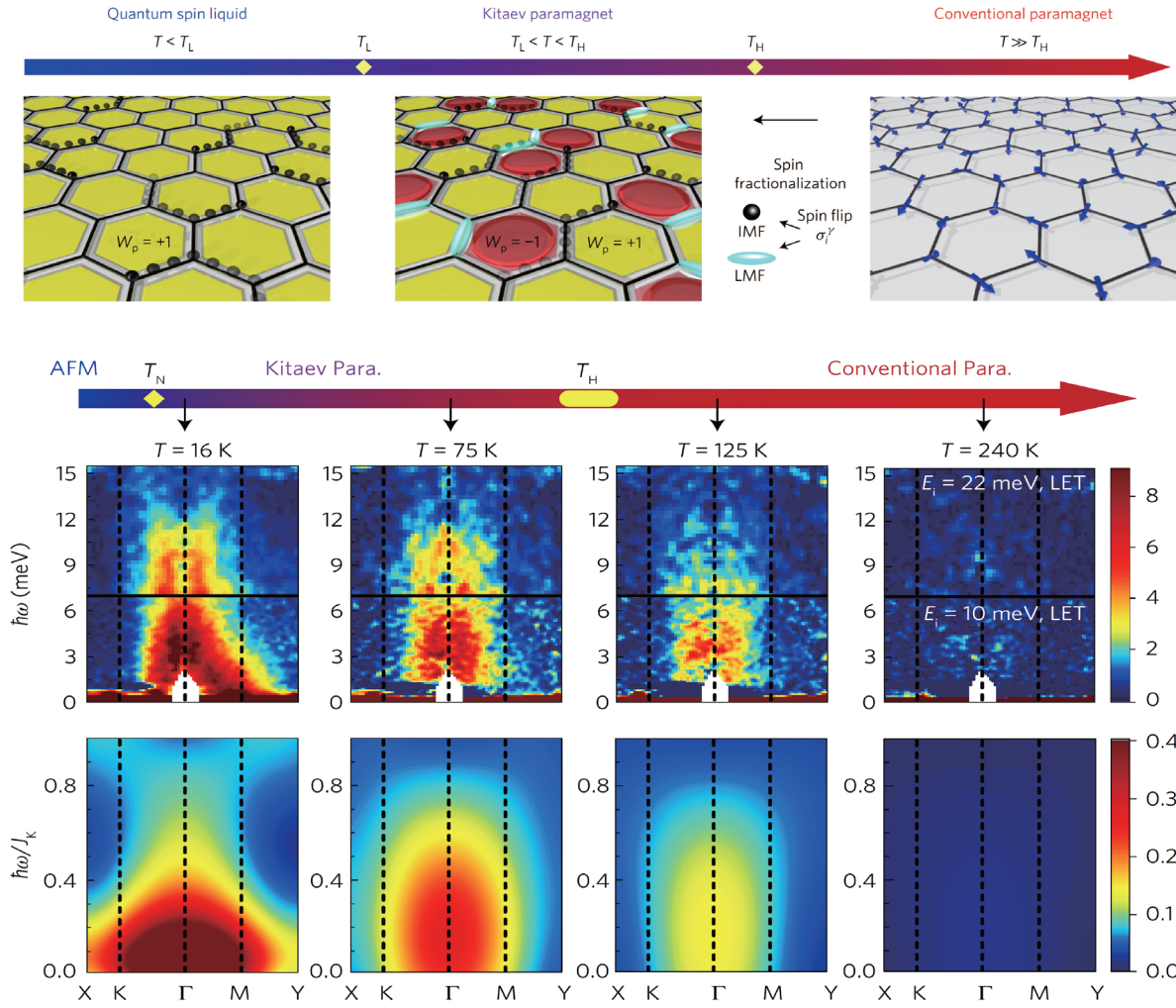


Figure 13. The upper panel sketches a thermal fractionalization of spins. In a Kitaev spin-liquid state, Z_2 -fluxes (yellow hexagon) are uniformly condensed and itinerant Majorana fermions (black balls) freely propagate. In the intermediate Kitaev paramagnetic state, the Z_2 -fluxes become thermally populated and the itinerant Majorana fermions are thermally damped. In the high- T paramagnetic state, only uncorrelated spins exist. The lower panel compares a thermal evolution of the experimental and calculated magnetic scattering function $S_{\text{mag}}(Q, \omega)$. [58] © 2017 Macmillan Publishers Limited. With permission of Springer.

order and the XY-like anisotropy. Needless to say, other non-Kitaev interactions should be taken into account to capture low-energy magnetic features.

The defining characteristic of the Kitaev QSL is an emergence of two fractional magnetic excitations. For the pure Kitaev model, $S_{\text{mag}}(Q, \omega)$ measured by INS comprises a coexistence of static flux excitation and odd numbers of itinerant Majorana fermions [8, 50]. Figure 13 compares the INS data of α -RuCl₃ with the simulated $S_{\text{mag}}(Q, \omega)$ as a function of temperature. Overall, the T dependence of the observed scattering continuum is reproduced by the simulated $S_{\text{mag}}(Q, \omega)$ for the pure FM Kitaev model with $K = 16.5$ meV [58]. The $T = 16$ K $S_{\text{mag}}(Q, \omega)$ shows an hour-glass shape spectrum centered at the Γ -point extending above 15 meV. The high-energy Y-shaped excitation corresponds to the dispersive itinerant Majorana fermion, while the quasielastic responses at low energies around the Γ -point are linked the flux excitations. With increasing temperature to $T = 75$ K (Kitaev paramagnetic phase), the low-energy intensity is substantially suppressed while the high-energy

intensity from the itinerant Majorana fermions is largely retained. Upon further heating towards a conventional paramagnetic phase, the dichotomic feature fades away due to thermal fluctuations and only a featureless background is visible. A close resemblance of the experimental and simulated spin excitations as a function of temperature supports the notion of thermal fractionalization of spins to Majorana fermions in α -RuCl₃ [58, 59].

However, it should be kept in mind that a discrepancy between the experiment and the simulation starts to appear in the temperature range below 50 K and for energies below 6 meV [58]. Especially, the hexagram-shaped Q -dependence of the low-energy $S_{\text{mag}}(Q, \omega)$ is explained by invoking the second nn Kitaev K' interactions and Γ interactions. This situation is reminiscent of the weakly coupled antiferromagnetic chain KCuF₃ that shows a crossover from a 1D Luttinger liquid to a 3D nonlinear physics (see section 3). In case of α -RuCl₃, the high-energy Kitaev paramagnet transits to the low-energy $K - \Gamma - K'$ model whose elementary excitations are different from Majorana fermions.

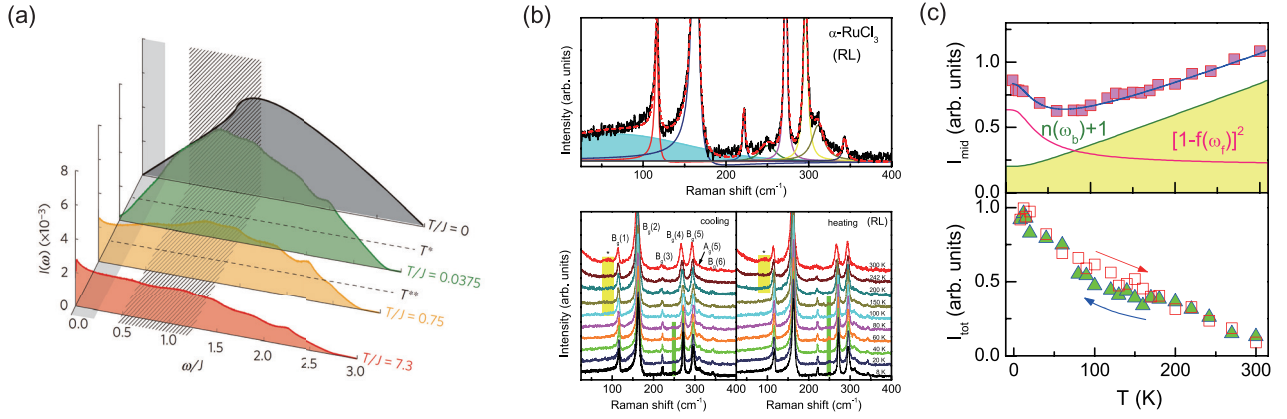


Figure 14. (a) Calculated Raman spectra of the pure Kitaev model as a function of temperature. Reprinted with permission from [10] © 2016 Macmillan Publishers Limited. (b) $T = 8$ K Raman spectrum of α -RuCl₃ fitted to a sum of a Gaussian profile (cyan shaded region), two Fano lines (red and blue solid lines), and six Lorentzian profiles (colored solid lines). T dependence of the Raman spectra measured in *RL* polarization is compared on cooling and heating. (c) (Upper panel) T dependence of the integrated Raman intensity $I_{\text{mid}}(\omega)$ obtained in the middle energy window of $5 < \hbar\omega < 14$ meV. The shaded area indicates the bosonic background and the solid red line is a fit to an expected T dependence for the two-fermionic process $[1 - f(\omega_f)]^2$. (Lower panel) Thermal hysteresis of the total intensity. Reprinted with permission from [60] © 2017 by the American Physical Society.

In contrast to neutron scattering, the Raman scattering process involves pairs of itinerant Majorana fermions [10, 51]. More significantly, Raman spectra of an ideal Kitaev model convey smoking-gun signatures of Majorana fermions. As shown in figure 14(a), the calculated $T = 0$ K spectrum displays asymptotically a ω -linear dependence in the low-energy region. This is related to a Dirac-like dispersion of itinerant Majorana fermions. With increasing temperature, the intermediate-energy spectral weight is gradually suppressed while the low-energy part increases. At high temperatures, the spectrum shows a featureless continuum. As discussed in section 2, the two distinct scattering processes, depending on the energy window, yield an asymptotic T dependence of the Raman intensity $I_{\text{mid}}(\omega) \propto [1 - f(\omega)]^2$ in the intermediate-energy range and $I_{\text{low}}(\omega) \propto f(\omega)[1 - f(\omega)]$ in the low-energy range.

Shown in figure 14(b) is the T dependence of the Raman spectra of α -RuCl₃ in circular (*RL*) polarization on heating and cooling [60]. We observe a broad magnetic continuum (marked by color shadings) for frequencies below 200 cm^{-1} (≈ 25 meV). The spectral bandwidth amounts to 1.51 K determined from the INS data. Compared to the calculated spectrum, we find a substantial low-energy residual spectral weight. Our $T = 8$ K spectrum bears a strong resemblance to the calculated high- T spectrum for the pure Kitaev model. This implies that non-Kitaev interactions behave like thermal damping, rendering the Kitaev QSL incoherent. As the temperature is raised, the low-energy part of the magnetic continuum evolves progressively into a quasielastic response. To shed light on the nature of the magnetic continuum, we calculate $I_{\text{mid}}(\omega)$ by integrating the magnetic Raman intensity over the middle energy range of $5 < \hbar\omega < 14$ meV and plot its T dependence in figure 14(c). It turns out that $I_{\text{mid}}(\omega)$ is described by a combination of the Bose factor $1 + n(\omega_b) = 1/[1 - \exp(-\hbar\omega_b/k_B T)]$ and the two-fermionic term $[1 - f(\omega_f)]^2$ with the effective energies of the bosonic and fermionic excitations $\hbar\omega_b = 5.8$ meV and $\hbar\omega_f = 5.0$

meV, respectively. Essentially the same results have been reported by other groups [10, 61]. We note in passing that the T dependence of the low-energy intensity $I_{\text{low}}(T)$ in the energy interval of $\hbar\omega = 1.5\text{--}7$ meV obeys neither the theoretically predicted $f(\omega)[1 - f(\omega)]$ dependence nor a combination of the bosonic and fermionic terms. Based on the analysis of our Raman spectra in terms of the Fermi statistics, we conclude that the continuum excitations contain the scattering contribution from the bosonic and fermionic quasiparticles for energies above 5 meV. However, the nature of the low-energy excitations could not be unambiguously identified in terms of the particle statistics. Unlike the INS data, a crossover from the high-energy Majorana to the low-energy magnon excitations is not apparent.

Some comment is in order on this issue. In the weakly coupled antiferromagnetic chain KCuF₃, a one-to-one correspondence between the INS and Raman scattering results is found in their energy dependence (compare figures 6(e) and 7(c)). This is related to the fact that both the neutron and Raman scattering are commonly governed by two spinons. In the case of a spin chain, we recall that spinons carrying $s = 1/2$ are created in pairs. Given that light has a very long wavelength compared to the unit cell and the light scattering involves the electric field and not the magnetic field, the Raman scattering process must have zero total spin momentum (see table 1). Thus, the Raman (neutron) scattering probes the singlet (triplet) sector of a two-spinon continuum. As to the Kitaev system, spins are fractionalized to fluxes and itinerant Majorana fermions with $s = 1/2$. Both excitations are directly probed by neutron scattering. However, the hexagonal loop operator defining the fluxes commutes with the Raman operator, thereby the flux excitations are hidden to the Raman scattering. Considering the selection rule, the Raman scattering process probes mainly two Majorana excitations whose spin components are opposite in their directions. As a consequence, a direct comparison of the neutron and Raman data is less probable for the Kitaev honeycomb case.

6. Outlook and conclusion

In this review, we have provided a comparative perspective of fractionalized spin excitations between neutron and Raman scatterings in QSL candidate materials. Three different classes of materials are chosen including the 1D antiferromagnetic chain KCuF_3 , the kagome antiferromagnet $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$, and the Kitaev honeycomb material $\alpha\text{-RuCl}_3$. Although these three compounds have different spin topology and magnetic exchange interactions, they share lots of similarity in magnetic Raman spectra. A continuum of magnetic excitations is commonly observed, which is taken as evidence for spin fractionalization. In addition, the spectral form and temperature dependence of the fractionalized continuum convey clues to particle statistics and the type of spin liquid.

We will close this review by giving an outlook of the challenges and possibilities that lie ahead for Raman spectroscopic studies of QSLs. As we established, Raman scattering can complement INS as an inexpensive, readily available tabletop method. In the future, it will be an indispensable tool to tackle unresolved issues such as a Landau level formation in strained thin films of Kitaev QSL materials [62], which would serve as a clear fingerprint of Majorana fermions with Dirac-like dispersion. Since Raman scattering experiments can be carried out in high magnetic fields up to 45 T, it is suitable for testing universal scaling laws expected to emerge from spin liquid states [63]. Furthermore, excitations from novel field-induced phases in Kitaev magnets [64] and related compounds can be investigated in detail, which reaches beyond the limits of inelastic neutron spectroscopy [65]. It also allows for studying the crossover from coupled layers of *quasi*-2D Kitaev honeycomb magnets to exfoliated *pure*-2D single layers to elucidate the effect of interlayer coupling on the Kitaev model. Finally, with the help of Raman spectroscopy, one can gain a deep understanding of the robustness and the fragility of QSLs as well as closely study the boundaries and the crossover to neighboring phases. This can be achieved by systematically doping Kitaev- and other QSL materials, which allows continuous tuning of the Hamiltonian of the system. All of these issues involve samples whose size and volume would not suffice for an INS study.

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References

- [1] Balents L 2010 Spin liquids in frustrated magnets *Nature* **464** 199
- [2] Zhou Y, Kanoda K and Ng T K 2017 Quantum spin liquid states *Rev. Mod. Phys.* **89** 025003
- [3] Knolle J and Moessner R 2019 A field guide to spin liquids *Annu. Rev. Condens. Matter Phys.* **10** 451
- [4] Ritchey I, Chandra P and Coleman P 1993 Spin folding in the two-dimensional Heisenberg kagomé antiferromagnet *Phys. Rev. B* **47** 15342
- [5] Lecheminant P, Bernu B and Lhuillier C 1997 Order versus disorder in the quantum Heisenberg antiferromagnet on the kagomé lattice using exact spectra analysis *Phys. Rev. B* **56** 2521
- [6] Liao H J, Xie Z Y, Chen J, Liu Z Y, Xie H D, Huang R Z, Normand B and Xiang T 2017 Gapless spin-liquid ground state in the $S = 1/2$ kagome antiferromagnet *Phys. Rev. Lett.* **118** 137202
- [7] Bethe H 1931 Zur Theorie der Metalle. I. Eigenwerte und Eigenfunktionen der linearen Atomkette *Z. Phys.* **71** 205
- [8] Kitaev A 2003 Fault-tolerant quantum computation by anyons *Ann. Phys.* **303** 2
- [9] Baskaran G, Mandal S and Shankar R 2007 Exact results for spin dynamics and fractionalization in the Kitaev model *Phys. Rev. Lett.* **98** 247201
- [10] Nasu J, Knolle J, Kovrizhin D L, Motome Y and Moessner R 2016 Fermionic response from fractionalization in an insulating two-dimensional magnet *Nat. Phys.* **12** 912
- [11] Aroyo M I, Perez-Mato J M, Orobengoa D, Tasci E, de la Flor G and Kirov A 2011 Crystallography online: bilbao crystallographic server *Bulg. Chem. Commun.* **43** 183
- Aroyo M I, Perez-Mato J M, Capillas C, Kroumova E, Ivantchev S, Madariaga G, Kirov A and Wondratschek H 2006 Bilbao crystallographic server I: databases and crystallographic computing programs *Z. Krist.* **221** 15
- Aroyo M I, Kirov A, Capillas C, Perez-Mato J M and Wondratschek H 2006 Bilbao crystallographic server II: representations of crystallographic point groups and space groups *Acta Crystallogr.* **A62** 115
- [12] Hayes W and Loudon R 1978 *Scattering of Light by Crystals* (New York: Wiley)
- [13] Fleury P A and Loudon R 1968 Scattering of light by one- and two-magnon excitations *Phys. Rev.* **166** 514
- [14] Elliott R J and Thorpe M F 1969 The effects of magnon-magnon interaction on the two-magnon spectra of antiferromagnets *J. Phys. C: Solid State Phys.* **2** 1630
- [15] Shastry B S and Shraiman B I 1990 Theory of Raman scattering in Mott-Hubbard systems *Phys. Rev. Lett.* **65** 1068
- [16] Arguello C A, Rousseau D L and Porto S P S 1969 First-order Raman effect in Wurtzite-type crystals *Phys. Rev.* **181** 1351
- [17] Yu P and Cardona M 2010 *Fundamentals of Semiconductors: Physics and Materials Properties* 4th edn (Berlin: Springer)
- [18] Choi K Y, Lemmens P, Heydhausen D, Güntherodt G, Baumann C, Klingeler R, Reutler P and Büchner B 2008 Anomalous orbital dynamics in LaSrMnO_4 observed by Raman spectroscopy *Phys. Rev. B* **77** 064415
- [19] Reiter G F 1976 Light scattering from energy fluctuations in magnetic insulators *Phys. Rev. B* **13** 169
- [20] Mermin N D and Wagner H 1966 Absence of ferromagnetism or antiferromagnetism in one- or two-dimensional isotropic Heisenberg models *Phys. Rev. Lett.* **17** 1133

- [21] Kadota S, Yamada I, Yoneyama S and Hirakawa K 1967 Formation of one-dimensional antiferromagnet in KCuF_3 with the perovskite structure *J. Phys. Soc. Japan* **23** 751
- [22] Deisenhofer J, Leonov I, Eremin M V, Kant Ch, Ghigna P, Mayr F, Iglamov V V, Anisimov V I and van der Marel D 2008 Optical evidence for symmetry changes above the Néel temperature of KCuF_3 *Phys. Rev. Lett.* **101** 157406
- [23] Lee J C T *et al* 2011 Two-stage orbital order and dynamical spin frustration in KCuF_3 *Nat. Phys.* **8** 63
- [24] Eremin M V, Zakharov D V, Krug von Nidda H V, Eremina R M, Shuvaev A, Pimenov A, Ghigna P, Deisenhofer J and Loidl A 2008 Dynamical Dzyaloshinsky–Moriya interaction in KCuF_3 *Phys. Rev. Lett.* **101** 147601
- [25] Gnezdilov V, Deisenhofer J, Lemmens P, Wulferding D, Afanasiev O, Ghigna P, Loidl A and Yermenko A 2012 Phononic and magnetic excitations in the quasi-one-dimensional Heisenberg antiferromagnet KCuF_3 *Low Temp. Phys.* **38** 419
- [26] des Cloizeaux J and Pearson J J 1962 Spin-Wave spectrum of the antiferromagnetic linear chain *Phys. Rev.* **128** 2131
- [27] Faddeev L D and Takhtajan L A 1981 What is the spin of a spin wave? *Phys. Lett. A* **85** 375
- [28] Tennant D A, Perring T G, Cowley R A and Nagler S E 1993 Unbound spinons in the $S = 1/2$ antiferromagnetic chain KCuF_3 *Phys. Rev. Lett.* **70** 4003
- [29] Lake B, Tennant D A, Frost C D and Nagler S E 2005 Quantum criticality and universal scaling of a quantum antiferromagnet *Nat. Mater.* **4** 329
- [30] Lake B, Tennant D A and Nagler S E 2005 Longitudinal magnetic dynamics and dimensional crossover in the quasi-one-dimensional spin-1/2 Heisenberg antiferromagnet KCuF_3 *Phys. Rev. B* **71** 134412
- [31] Yamada I and Onda H 1994 Light scattering from magnetic-energy fluctuations in the one-dimensional Heisenberg antiferromagnet KCuF_3 *Phys. Rev. B* **49** 1048
- [32] Helton J S *et al* 2007 Spin dynamics of the Spin-1/2 kagome lattice antiferromagnet $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$ *Phys. Rev. Lett.* **98** 107204
- [33] Huse D A and Rutenberg A D 1992 Classical antiferromagnets on the kagomé lattice *Phys. Rev. B* **45** 7536(R)
- [34] Yan S, Huse D A and White S R 2011 Spin-Liquid Ground State of the $S = 1/2$ kagome Heisenberg antiferromagnet *Science* **332** 1173
- [35] Hiroi Z, Yoshida H, Okamoto Y and Takigawa M 2009 Spin-1/2 kagome compounds: volborthite versus Herbertsmithite *J. Phys.: Conf. Ser.* **145** 012002
- [36] Han T H, Helton J S, Chu S, Nocera D G, Rodriguez-Rivera J A, Broholm C and Lee Y S 2012 Fractionalized excitations in the spin-liquid state of a kagome-lattice antiferromagnet *Nature* **492** 406
- [37] Ko W H, Liu Z X, Ng T K and Lee P A 2010 Raman signature of the U(1) Dirac spin-liquid state in the spin-1/2 kagome system *Phys. Rev. B* **81** 024414
- [38] Wulferding D, Lemmens P, Scheib P, Röder J, Mendels P, Chu S, Han T and Lee Y S 2010 Interplay of thermal and quantum spin fluctuations in the kagome lattice compound herbertsmithite *Phys. Rev. B* **82** 144412
- [39] de Vries M A, Wulferding D, Lemmens P, Lord J S, Harrison A, Bonville P, Bert F and Mendels P 2012 Extension of the zinc paratacamite phase diagram: probing the effect of spin vacancies in an $S = 1/2$ kagome antiferromagnet *Phys. Rev. B* **85** 014422
- [40] Wulferding D, Lemmens P, Yoshida H, Okamoto Y and Hiroi Z 2012 The spin dynamics in distorted kagome lattices: a comparative Raman study *J. Phys.: Condens. Matter* **24** 185602
- [41] Colman R H, Bert F, Boldrin D, Hillier A D, Manuel P, Mendels P and Wills A S 2011 Spin dynamics in the $S = 1/2$ quantum kagome compound vesignieite, $\text{Cu}_3\text{Ba}(\text{VO}_5\text{H})_2$ *Phys. Rev. B* **83** 180416
- [42] Okamoto Y, Yoshida H and Hiroi Z 2009 Vesignieite $\text{BaCu}_3\text{V}_2\text{O}_8(\text{OH})_2$ as a candidate spin-1/2 kagome antiferromagnet *J. Phys. Soc. Japan* **78** 033701
- [43] Sindzingre P 2007 Thermodynamic properties of the spin-1/2 Heisenberg antiferromagnet with anisotropic exchange on the kagome lattice: comparison with volborthite (arXiv:0707.4264)
- [44] Nilsen G J, Coomer F C, de Vries M A, Stewart J R, Deen P P, Harrison A and Ronnow H M 2011 Pair correlations, short-range order, and dispersive excitations in the quasi-kagome quantum magnet volborthite *Phys. Rev. B* **84** 172401
- [45] Kohama Y, Ishikawa H, Matsuo A, Kindo K, Shannon N and Hiroi Z 2019 Possible observation of quantum spin-nematic phase in a frustrated magnet *Proc. Natl Acad. Sci.* **116** 10686
- [46] Rau J G, Lee E K H and Kee H Y 2016 Spin–orbit physics giving rise to novel phases in correlated systems: iridates and related materials *Annu. Rev. Condens. Matter Phys.* **7** 195
- [47] Jackeli G and Khaliullin G 2009 Mott insulators in the strong spin–orbit coupling limit: from Heisenberg to a quantum compass and Kitaev Models *Phys. Rev. Lett.* **102** 017205
- [48] Singh Y and Gegenwart P 2010 Antiferromagnetic Mott insulating state in single crystals of the honeycomb lattice material Na_2IrO_3 *Phys. Rev. B* **82** 064412
- [49] Plumb K W, Clancy J P, Sandilands L J, Shankar V V, Hu Y F, Burch K S, Kee H Y and Kim Y J 2014 $\alpha\text{-RuCl}_3$: a spin–orbit assisted Mott insulator on a honeycomb lattice *Phys. Rev. B* **90** 041112
- [50] Knolle J, Kovrizhin D L, Chalker J T and Moessner R 2014 Dynamics of a two-dimensional quantum spin liquid: signatures of emergent majorana fermions and fluxes *Phys. Rev. Lett.* **112** 207203
- [51] Knolle J, Chern G W, Kovrizhin D L, Moessner R and Perkins N B 2014 Raman scattering signatures of Kitaev spin liquid in A_2IrO_3 iridates with $A = \text{Na}$ or Li *Phys. Rev. Lett.* **113** 187201
- [52] Nasu J, Udagawa M and Motome Y 2015 Thermal fractionalization of quantum spins in a Kitaev model: temperature-linear specific heat and coherent transport of Majorana fermions *Phys. Rev. B* **92** 115122
- [53] Nasu J, Yoshitake J and Motome Y 2017 Thermal transport in the Kitaev model *Phys. Rev. Lett.* **119** 127204
- [54] Glamazda A, Lemmens P, Do S-H, Choi Y S and Choi K Y 2016 Raman spectroscopic signature of fractionalized excitations in the harmonic-honeycomb iridates β - and γ - Li_2IrO_3 *Nat. Commun.* **7** 12286
- [55] Gupta S N, Sriluckshmy P V, Mehlatat K, Balodhi A, Mishra D K, Hassan S R, Ramakrishnan T V, Muthu D V S, Singh Y and Sood A K 2016 Raman signatures of strong Kitaev exchange correlations in $(\text{Na}_{1-x}\text{Li}_x)_2\text{IrO}_3$: experiments and theory *Europhys. Lett.* **114** 47004
- [56] Gupta S N, Sriluckshmy P V, Balodhi A, Muthu D V S, Hassan S R, Singh Y, Ramakrishnan T V and Sood A K 2016 Spin liquid like Raman signatures in the hyperkagome iridate $\text{Na}_4\text{Ir}_3\text{O}_8$ *Phys. Rev. B* **94** 155153
- [57] Park S Y *et al* 2016 Emergence of the isotropic Kitaev honeycomb lattice with two-dimensional Ising universality in $\alpha\text{-RuCl}_3$ (arXiv:1609.05690)
- [58] Do S H *et al* 2017 Majorana fermions in the Kitaev quantum spin system $\alpha\text{-RuCl}_3$ *Nat. Phys.* **13** 1079
- [59] Banerjee A, Yan J, Knolle J, Bridges C A, Stone M B, Lumsden M D, Mandrus D G, Tennant D A, Moessner R and Nagler S E 2017 Neutron scattering in the proximate quantum spin liquid $\alpha\text{-RuCl}_3$ *Science* **356** 1055
- [60] Glamazda A, Lemmens P, Do S H, Kwon Y S and Choi K Y 2017 Relation between Kitaev magnetism and structure in $\alpha\text{-RuCl}_3$ *Phys. Rev. B* **95** 174429

- [61] Sandilands L J, Tian Y, Plumb K W, Kim Y J and Burch K S 2015 Scattering continuum and possible fractionalized excitations in α -RuCl₃ *Phys. Rev. Lett.* **114** 147201
- [62] Perreault B, Rachel S, Burnell F J and Knolle J 2017 Majorana Landau-level Raman spectroscopy *Phys. Rev. B* **95** 184429
- [63] Modic K A *et al* 2019 Scale-invariance of a spin liquid in high magnetic fields (arXiv:1901.09245)
- [64] Kasahara Y *et al* 2018 Majorana quantization and half-integer thermal quantum Hall effect in a Kitaev spin liquid *Nature* **559** 227
- [65] Banerjee A *et al* 2018 Excitations in the field-induced quantum spin liquid state of α -RuCl₃ *NPJ Quantum Mater.* **3** 8